

Multiple Pattern Search

Topic.

String algorithms, trie, multiple pattern matching.

Statement.

Given a text and a dictionary of words.

For each dictionary word, determine whether it appears in the text as a substring.

Input.

The first line contains the text. Its length is at most

$$10^6.$$

The text consists of lowercase English letters.

The next line contains an integer

$$M,$$

the number of words in the dictionary.

Each of the next M lines contains one dictionary word. Every word has length at most 30.

All words are distinct and sorted in lexicographic order.

The total length of all dictionary words is at most

$$10^5.$$

Output.

Print M lines.

For each dictionary word, print

Yes

if the word occurs in the text as a substring, and

No

otherwise.

Implementation.

The implementation is provided separately in the file

`solution.cpp`.